

## A2 Practice 7.1 and 7.2 v2

1. Given the following exponential function, identify whether the change represents growth or decay, and determine the percentage rate of increase or decrease.

$$y = 9100(0.92)^x$$

$.92 = \text{Decay}$  b/c  $< 1$   
 $.92 = 1 - r$  b/c decay  
 $-.08 = -r$   
 $r = .08 \rightarrow \boxed{8\% \downarrow}$

2. The function  $f(t) = 4100(0.9955)^{60t}$  represents the change in a quantity over  $t$  hours. What does the constant 0.9955 reveal about the rate of change of the quantity?

The function is decaying exponentially at a rate of 45 % every minute  
word bank 1  
word bank 2

Word bank 1: (a) decaying, (b) growing

Word bank 2: (a) day, (b) hour, (c) 60 days, (d) 60 hours, (e) minute

$$.9955 = \text{decay}$$

$$.9955 = 1 - r$$

$$-.0045 = -r$$

$$r = .0045 \rightarrow 45\%$$


---

time = 1 hour  
 $60t = 1 \text{ hour}$   
 $t = \frac{1}{60} \text{ hour} = 1 \text{ minute}$

3. A new car is purchased for 17300 dollars. The value of the car depreciates at 9.25% per year. What will the value of the car be, to the nearest cent, after 15 years?

decreases  $9.25\% = .0925$

$$A = 17300(1 - .0925)^{15}$$

$$A = 4034.090239$$

$\uparrow$

$$A = \boxed{4,034.09}$$

4. Write an exponential function in the form  $y = ab^x$  that goes through the points  $(0, 15)$  and  $(3, 960)$ .

$\uparrow$   
 $a$   
 $y = ab^x$  general formula  
 $\Rightarrow$  need to find specific  $a$  and  $b$

$$y = 15b^x$$

$$960 = 15b^3$$

$$64 = b^3$$

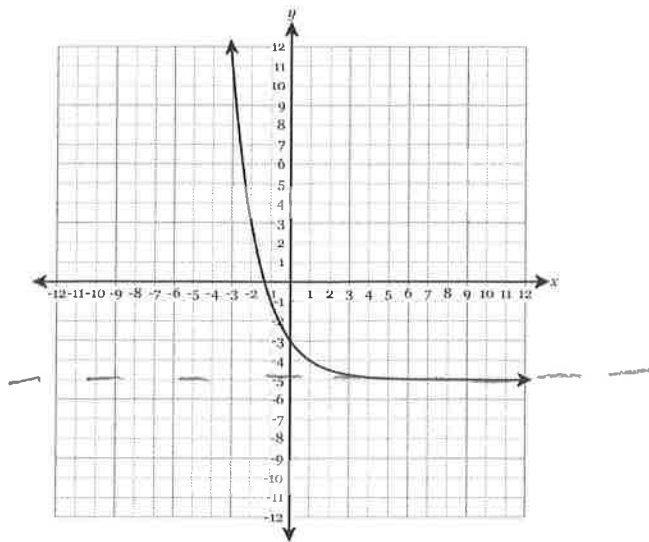
$$\sqrt[3]{b^3} = \sqrt[3]{64}$$

$$b = 4 \text{ (just the positive)}$$

$\Rightarrow$  Answer the question!

$$\boxed{y = 15(4)^x}$$

5. What are the features of the function  $f(x) = 2\left(\frac{1}{2}\right)^x - 5$  graphed below?



The function  $f(x)$  is exponential function with a horizontal asymptote of  $y = -5$ . The range of the function is  $(-5, \infty)$ , and it is decreasing on its domain of  $(-\infty, \infty)$ . The end behavior on the LEFT side is as  $x \rightarrow -\infty$ ,  $y \rightarrow \infty$ , and the end behavior on the RIGHT side is as  $x \rightarrow \infty$ ,  $y \rightarrow -5$ .

Word bank 1: (a) a logarithmic, (b) an exponential

Word bank 2: (a) horizontal, (b) vertical

Word bank 3: (a)  $x =$ , (b)  $y =$

Word bank 4: (a)  $(-\infty, 0)$ , (b)  $(-\infty, \infty)$ , (c)  $(-\infty, -5)$ , (d)  $(-5, \infty)$ , (e)  $(5, \infty)$ , (f)  $(0, \infty)$ , (g)  $(-\infty, 5)$

Word bank 5: (a) increasing, (b) decreasing

Word bank 6: (a)  $(0, \infty)$ , (b)  $(5, \infty)$ , (c)  $(-\infty, -5)$ , (d)  $(-\infty, \infty)$ , (e)  $(-\infty, 0)$ , (f)  $(-5, \infty)$ , (g)  $(-\infty, 5)$

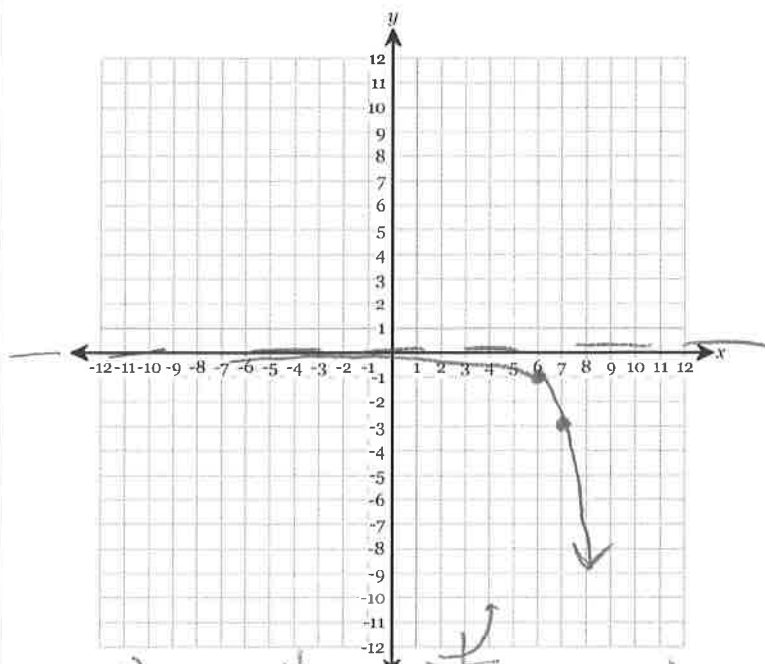
Word bank 7: (a)  $x \rightarrow -5$  (from the right), (b)  $x \rightarrow -\infty$

Word bank 8: (a)  $y \rightarrow -5$ , (b)  $y \rightarrow \infty$ , (c)  $y \rightarrow -\infty$

Word bank 9: (a)  $x \rightarrow -5$  (from the left), (b)  $x \rightarrow \infty$

Word bank 10: (a)  $y \rightarrow \infty$ , (b)  $y \rightarrow -5$ , (c)  $y \rightarrow -\infty$

6. Graph the function  $f(x) = -3^{x-6}$  on the axes below. You must plot the asymptote and any two points with integer coordinates.



- 1) growth
- 2) flipped over x axis
- 3) move right 6 units
- 4) asymptote @  $y = 0$
- 5) Find 2 easy points

$$\text{Exp} = 0$$

$$x - 6 = 0$$

$$x = 6$$

SUB BACK

$$f(6) = -3^{6-6} = -3^0 = -1$$

$$(6, -1)$$

$$\text{Exp} = 1$$

$$x - 6 = 1$$

$$x = 7$$

SUB BACK

$$f(7) = -3^{7-6} = -3^1 = -3$$

$$(7, -3)$$